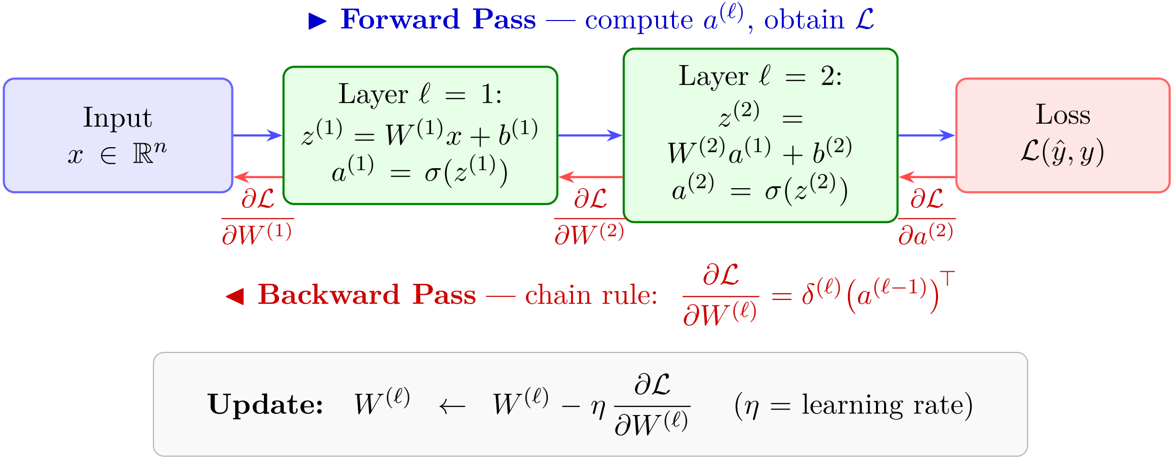
▶ **Forward Pass** — compute $a^{(\ell)}$, obtain $\mathcal{L}$
Input $x \in \mathbb{R}^n$
Layer $\ell = 1$: $z^{(1)} = W^{(1)}x + b^{(1)}$ $a^{(1)} = \sigma(z^{(1)})$
Layer $\ell = 2$: $z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$ $a^{(2)} = \sigma(z^{(2)})$
Loss $\mathcal{L}(\hat{y}, y)$
$\frac{\partial \mathcal{L}}{\partial W^{(1)}}$
$\frac{\partial \mathcal{L}}{\partial W^{(2)}}$
$\frac{\partial \mathcal{L}}{\partial a^{(2)}}$
◀ **Backward Pass** — chain rule: $\frac{\partial \mathcal{L}}{\partial W^{(\ell)}} = \delta^{(\ell)} \left(a^{(\ell-1)}\right)^{\top}$
Update: $W^{(\ell)} \leftarrow W^{(\ell)} - \eta \frac{\partial \mathcal{L}}{\partial W^{(\ell)}}$ (η = learning rate)